MOOC Econometrics — Test Exercise 6 Full Solution Guide

Dataset Context: Monthly CPI Data for Euro Area (EA) & United States (USA) (Jan 2000 - Dec 2011)

Part (a): Time Series Visual Analysis & Stationarity Assessment

Action Required: Plot the raw CPI series of the Euro Area (EA) and the USA, their logarithm transformations (log(CPI)), and their monthly inflation rates defined as Δlog(CPI).

Econometric Analysis:
1. **Raw CPI Levels:** Show a strong, continuous deterministic upward trend. This confirms the presence of non-stationarity in levels.
2. **Log(CPI) Series:** Linearizes the exponential growth, but the series remains strongly trended and non-stationary.
3. **First Differences (Δlog(CPI)):** The transformation successfully eliminates the long-run deterministic trend. The resulting monthly inflation series fluctuates around a stable mean, showing characteristics of a stationary series.

Part (b): Augmented Dickey-Fuller (ADF) Test Specification

Test Equation: To test for a unit root in log(CPI) series while accounting for trend and serial correlation, we estimate the following ADF specification with a constant, a linear trend, and 3 lags:

$$\Delta y_t = \alpha + \beta\, t + \gamma\, y_{t-1} + \delta_1\, \Delta y_{t-1} + \delta_2\, \Delta y_{t-2} + \delta_3\, \Delta y_{t-3} + \varepsilon_t$$

Hypothesis Testing:
- **Null Hypothesis (H_0):** $\gamma = 0$ (The series contains a unit root / is non-stationary).
- **Alternative Hypothesis (H_1):** $\gamma < 0$ (The series is trend-stationary).

Interpretation Strategy: The t-statistic computed for γ is compared against the MacKinnon critical values for a model with a constant and trend. If the t-statistic is more negative than the critical value, reject H_0. For levels, we fail to reject H_0, confirming that log(CPI) contains a unit root.

Part (c): Autoregressive AR(2) Baseline Model for EA Inflation

Model Specification: Under the assumption that the log(CPI) series are not cointegrated, we model the Euro Area monthly inflation series ($INF_{EA,t} = \Delta\log(CPI_{EA,t})$) using an AR(2) model over the estimation period **January 2000 to December 2010**:

$$INF_{EA,t} = \mu + \phi_1\, INF_{EA,t-1} + \phi_2\, INF_{EA,t-2} + \varepsilon_t$$

Estimation Protocol: Ordinary Least Squares (OLS) estimation will yield coefficients μ, ϕ_1, and ϕ_2 along with their respective standard errors, t-values, and p-values. These parameters capture the short-run momentum and autoregressive dependency of inflation in the Euro Area.

Part (d): Expanded Autoregressive Distributed Lag (ADL) Model

Model Specification: We extend the baseline AR(2) model by incorporating lagged values of the US monthly inflation rate ($INF_{USA,t} = \Delta\log(CPI_{USA,t})$) at specific lags (1, 6, and 12):
$INF_{EA,t} = \mu + \phi_1 INF_{EA,t-1} + \phi_2 INF_{EA,t-2} + \beta_1 INF_{USA,t-1} + \beta_6 INF_{USA,t-6} + \beta_{12} INF_{USA,t-12} + \eta_t$

Significance Evaluation: We perform an explicit t-test on the coefficient of the 6th lag (β_6). If its p-value is greater than 0.05, we conclude that the 6th lag is statistically insignificant. A joint F-test can be conducted to evaluate whether US inflation Granger-causes Euro Area inflation.

Part (e): Out-of-Sample Rolling Forecasts & Performance Evaluation

Methodology: Generate 12 one-step-ahead monthly rolling forecasts for the hold-out year **2011**. For each month *m* in 2011, re-estimate the models using all available data up to month *m-1*, then forecast the inflation for month *m*.

Evaluation Metrics Formulations:

Metric Name	Mathematical Formula	Interpretation
RMSE Root Mean Squared Error	$\sqrt{[\ (1/N)\ \sum\ (e_{t+h})^2\ ]}$	Penalizes larger forecast errors heavily.
MAE Mean Absolute Error	$(1/N)\ \sum\ \lvert e_{t+h} \rvert$	Measures average absolute forecast deviation.
SUM Sum of Forecast Errors	$\sum\ e_{t+h}$	Indicates systematic forecast bias (over/under estimation).

Model Selection Rule: The model that yields the lowest absolute values for both RMSE and MAE, and a SUM closest to zero, is selected as the superior predictive model.

Executable R Code to Process the Provided Dataset File

Save your dataset text file as TestExer6-CPI-round1.txt in your working directory and execute the following complete econometric script to generate all outputs and figures:

```
# --- MOOC Econometrics Test Exercise 6 Full Automation Script ---
library(urca)
library(lmtest)

# 1. Load Dataset
data <- read.table("TestExer6-CPI-round1.txt", header=TRUE, stringsAsFactors=FALSE)
colnames(data)[1] <- "Year_Month"

# Create Time Series Objects
cpi_ea  <- ts(data$CPI_EUR, start=c(2000, 1), frequency=12)
cpi_usa <- ts(data$CPI_USA, start=c(2000, 1), frequency=12)

# Transformations
log_cpi_ea  <- log(cpi_ea)
log_cpi_usa <- log(cpi_usa)
inf_ea  <- diff(log_cpi_ea)
inf_usa <- diff(log_cpi_usa)
```

```
# 2. Part (b): Augmented Dickey-Fuller Test (Lag = 3, with Constant and Trend)
adf_ea  <- ur.df(log_cpi_ea, type="trend", lags=3)
adf_usa <- ur.df(log_cpi_usa, type="trend", lags=3)
print(summary(adf_ea))
print(summary(adf_usa))

# 3. Align Data Matrix for Regressions (Sample: Jan 2000 - Dec 2010)
full_df <- data.frame(
  inf_ea  = as.numeric(inf_ea),
  inf_usa = as.numeric(inf_usa)
)
# Add Lags
full_df$inf_ea_l1 <- c(NA, full_df$inf_ea[-nrow(full_df)])
full_df$inf_ea_l2 <- c(NA, NA, full_df$inf_ea[-c(nrow(full_df)-1, nrow(full_df))])

full_df$inf_usa_l1  <- c(NA, full_df$inf_usa[-nrow(full_df)])
full_df$inf_usa_l6  <- dplyr::lag(full_df$inf_usa, 6)
full_df$inf_usa_l12 <- dplyr::lag(full_df$inf_usa, 12)

# Filter Estimation Sample (Up to Dec 2010 -> index 1 to 131)
est_sample <- full_df[1:131, ]

# Part (c): AR(2) Model Estimation
model_c <- lm(inf_ea ~ inf_ea_l1 + inf_ea_l2, data=est_sample)
summary(model_c)

# Part (d): ADL Model Estimation
model_d <- lm(inf_ea ~ inf_ea_l1 + inf_ea_l2 + inf_usa_l1 + inf_usa_l6 + inf_usa_l12, data=est
_sample)
summary(model_d)

# 4. Part (e): 2011 Rolling Out-of-Sample Forecasts Execution
actual_2011 <- full_df[132:143, "inf_ea"]
fc_errors_c <- numeric(12)
fc_errors_d <- numeric(12)

for(i in 1:12) {
  current_end <- 131 + i - 1
  train_data  <- full_df[1:current_end, ]
  test_row    <- full_df[current_end + 1, ]

  # Re-estimate Models
  m_c_roll <- lm(inf_ea ~ inf_ea_l1 + inf_ea_l2, data=train_data)
  m_d_roll <- lm(inf_ea ~ inf_ea_l1 + inf_ea_l2 + inf_usa_l1 + inf_usa_l6 + inf_usa_l12, data=
train_data)

  # Forecast
  pred_c <- predict(m_c_roll, newdata=test_row)
  pred_d <- predict(m_d_roll, newdata=test_row)

  # Errors
  fc_errors_c[i] <- test_row$inf_ea - pred_c
  fc_errors_d[i] <- test_row$inf_ea - pred_d
}

# Metrics Calculation Function
```

```
calc_metrics <- function(errors) {
  rmse <- sqrt(mean(errors^2))
  mae  <- mean(abs(errors))
  sum_err <- sum(errors)
  return(c(RMSE=rmse, MAE=mae, SUM=sum_err))
}

cat("\n--- Forecast Performance Metrics (2011 Hold-out Sample) ---\n")
cat("AR(2) Model:\n"); print(calc_metrics(fc_errors_c))
cat("ADL Model:\n"); print(calc_metrics(fc_errors_d))
```

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